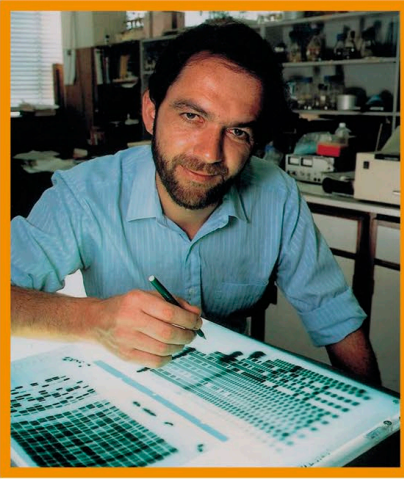


See also: INTERPOL 220–221 ■ The Universal Declaration of Human Rights 222–229 ■ The European Convention on Human Rights 230–233 ■ The International Convention Against Doping in Sport 304



guilt. In 1986, Richard Buckland, a youth with learning difficulties, was arrested for the rape and murder of 15-year-old Dawn Ashworth in the UK. The case came to the attention of British geneticist Alec Jeffreys. He examined DNA taken from semen found on Ashworth's corpse and showed that it was not Buckland's. Jeffreys also demonstrated that whoever killed Ashworth had killed another girl, Lynda Mann, 3 years earlier. Buckland was released from custody, and after a long investigation, serial sex offender Colin Pitchfork was found to have a DNA match with the semen samples. He was found guilty on two counts of murder and sentenced to life imprisonment.

Advances in DNA testing

Quick to exploit DNA as a major advance, law enforcement agencies around the world set up databases to store and share information. The US Federal Bureau of Investigation (FBI) launched its Combined DNA Index System in 1998; by 2020, it held DNA profiles of more than 14 million offenders, gathered from

Alec Jeffreys carried out research on genetics at Leicester University, in the UK, from 1977 to 2012. His work has been useful in paternity and immigration disputes, as well as in crime detection.

crime scenes. Similarly, the UK's National DNA Database (NDNAD), created in 1995, had 6.5 million profiles by 2020.

Since DNA profiling was first used, advances in technology have made it possible to analyze tiny samples very quickly. The clear success of DNA profiling has led to an assumption that it is infallible, but human error has led to several miscarriages of justice. In 1998, for example, 16-year-old Josiah Sutton from Houston, Texas, was given a 25-year jail sentence for rape based on DNA evidence. This was later shown to have been mishandled in the forensics laboratory, and Sutton was exonerated.

Shortcomings

In reality, as the number of stored DNA samples increases, the risk of blunders—of files muddled or

“People have unrealistic perceptions of the meaning of scientific evidence, especially when it comes to DNA.

EUROFORGEN

(European Forensic Genetics Network of Excellence), 2017

testing techniques misapplied—also grows. “Secondary transfer,” where DNA is transferred to an object or person through an intermediate, has proved to be a difficulty. Also, most everyday objects carry tiny traces of multiple people, and it can be almost impossible to distinguish the DNA of those in frequent contact. As a UK government report explained, “Our ability to analyze [DNA] may outstrip our ability to interpret.” ■

The Green River Killer

The Green River Killer serial murder case—one of the worst in US history—spectacularly vindicated the use of DNA profiling. Gary Ridgway raped and strangled at least 49 young women in the state of Washington between 1982 and 2001, when he was finally arrested. He dumped their bodies in or near the Green River. The police had no leads.

In 2001, improved DNA sampling techniques led to a crucial breakthrough for the

police investigation. DNA samples taken from semen found on some of the victims were compared with one taken from Ridgway in 1987, when, suspected of the killings, he had been interviewed but not charged. The samples proved to be an exact match; Ridgway was found guilty of murder, and he was sentenced to 48 consecutive life sentences. He later claimed to have killed as many as 80 women, and the true total may have been more.